

Instructions: There are totally 1 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Let $f(x) = x^4 - 12x^3 + 2$.

(a) Find all critical points of f . $\rightarrow f$ is differentiable everywhere.

$$f'(x) = 4x^3 - 36x^2$$

$$= 4x^2(x - 9)$$

$$x = 0 \text{ or } x = 9.$$

Thus, we have two critical points, 0 and 9.

(b) Find the intervals on which f is increasing and the intervals on which f is decreasing.

$$-\infty \quad 0 \quad 9 \quad +\infty$$

Intervals $f'(x)$

$$x < 0$$

$$-$$

$f(x)$ is decreasing on $(-\infty, 0]$

$$0 < x < 9$$

$$-$$

$f(x)$ is decreasing on $[0, 9]$

$$x > 9$$

$$+$$

$f(x)$ is increasing on $[9, +\infty)$

(c) Find the intervals on which f is concave up and the intervals on which f is concave down.

$$f''(x) = 12x^2 - 72x$$

$$= 12x(x - 6)$$

$$-\infty \quad 0 \quad 6 \quad +\infty$$

Intervals $f''(x)$

$$x < 0$$

$$+$$

$f(x)$ is concave up on $(-\infty, 0)$

$$0 < x < 6$$

$$-$$

$f(x)$ is concave down on $(0, 6)$

$$x > 6$$

$$+$$

$f(x)$ is concave up on $(6, +\infty)$

(d) Find the x -coordinates of all inflection points.

$$f''(x) = 0$$

$$12x(x - 6) = 0$$

$$x = 0 \text{ or } x = 6.$$

Thus, f has an inflection point at $x = 0$ and $x = 6$ respectively.